Expt: 08 Nmap Using Tiny Hack Me
Date: 01/10/2026

AIM:

To discover live host on a network using Nmap host discovery scans - ARP scan, IMP scan and TCP/UDP P Scan and to understand which protocol operates on w Layer of TCP/IP Model and when each one applies

ARP host Discovery

$ sudo nmap -PR -Sn 10.200.6.0/24

OUTPUT

Starting NMap 7.80 (https://nmap.org)
Nmap Scan report for 10.200.6.1
Host is up
Nmap done : 256 IP address (1 host up) scanned

ICMP Host Discovery

ICMP Echo (-PE)

$ sudo nmap -PE -Sn 10.200.6.0/24
nmap Scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap Scan report for 10.200.6.50.
Host is up (0.070 latency)
Nmap Scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done : 256 IP addresses (3 hosts up)
Scanned in 3.38 seconds

ICMP Timestamp (-PP)

$ nmap -PP -Sn 10.200.6.0/24
Nmap Scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap Scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done : 256 IP address (3 hosts up)
in 8.44 seconds

ICMP

$nmap -PM -sn 10.200.6.0/24

Nmap done: 256 IP address (0 hosts up) scanned in 52.14 seconds

TCP UDP Ping Discovery

i) TCP SYN Ping (-PS)

$nmap -PS -sn 10.200.6.0/24

Nmap Scan report for 10.200.6.50

Host is up (0.07s latency)

Nmap scan report for 10.200.6.250

ii) TCP ACK Ping (-PA)

Sends a TCP Packet with the ACK flag set

$sudo nmap - PA -sn 10.200.6.0/24

Nmap Scan report for 10.200.6.50

Host is up (0.070s latency)

Nmap Scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done: 256 IP address (2 hosts up) scanned in 36.64 seconds

iii) UDP Ping

$ sudo nmap - PU -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0070 latency)

Nmap scan report for 10.200.6.250

Host is up (0.069 latency)

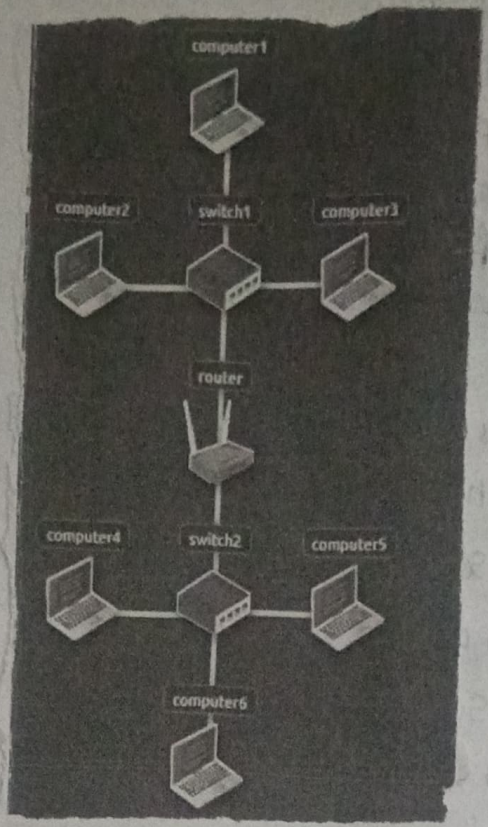
Nmap done : 256 IP address (2 hosts up) scanned in 35.97s

Masscan - A Router Alternative:

$masscan 10.200.6.0/24 - P443

$ masscan 10.200.6.0/24 - P80.443

$ masscan 10.200.6.0/24 - P22-25



Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

Correct Answer

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

Correct Answer

Which TCP ping scan does not require a privileged account?

Correct Answer

Which TCP ping scan requires a privileged account?

Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

Correct Answer

RESULT:

Thus the host discovery for ARP, ICMP TCP UDP was studied and the output is verified Successfully